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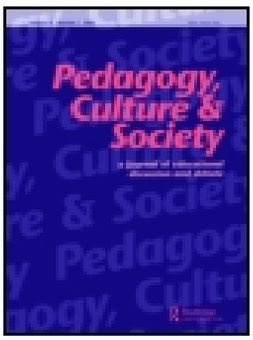
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Imagining imagination: towards cognitive and metacognitive models

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ABSTRACT

This paper presents a theoretical exploration of the relationship between imagination, cognition and metacognition, conceptualised within 'emergent models'. These models are offered to enable dialogue and tools to understand and support imagination in education practice, through the presence of ever-transforming theory, conceived as emergent phenomena. The models are arrived at theoretically and pragmatically, drawing on evolutionary, philosophical, psychological and art paradigms. They demonstrate a fundamental relationship between cognition, metacognition and imagination. A major conclusion is that to nurture thinking, we should perceive and develop imagination as a fundamental aspect of thought, as opposed to often regarding it as a magical force. Of special importance is the necessity for and prominence of imagination in metacognition, given the strong evidence that metacognition supports achievement and attainment. Through its fundamental relationship with metacognition, imagination increases our capacity to learn, enables personal and potentially, democratic capacity.

KEYWORDS

Imagination; metacognition; creativity; emergence; creative imagination; fantastical imagination; imagery

Introduction

There is ongoing international interest in creativity, creative practices and creative pedagogies. Creativity is featured as a domain in the 2022 PISA survey and is a key competency within the US-based Partnership for 21st Century Skills (P21) framework for C21 skills (2009). National (UK) initiatives have included the English, Creative Partnerships programme (2002–2011), with the Durham Commission on Creativity and Education (Arts Council England and Durham University 2019), Education Scotland (2013) and the Welsh Government (2019) recognising the need to support creativity through schools. Much of the theoretical work associated with these programmes has related to understanding the nature of creativity as applied to educational settings (e.g., Craft 2001; Kaufman and Beghetto 2009; Lucas 2019). This paper offers a complementary and less discussed question, that of imagination and its association with cognition and metacognition.

The author presents 'emergent models' of imagination, describing its relationship with cognition and metacognition. As Ryle (1949, 232) states 'Operations of imagining are, of course, exercises of mental powers'. Imagination is defined, initially, via Hume ([1739]

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1985), as "the capacity to think about things in their absence" (in Warnock 1976, 15) based on sensed "impressions" which become "ideas" (Hume [1739] 1985, Book 1, II). Cognition is defined as the mental process involved in knowing, learning, and understanding things' (Collins English Dictionary). It is understood as incorporating faculties of volition and emotion, enabling a comprehensive exploration of imagination's role within our thinking and overcoming limitations in situating it within a more restricted classification (see Ryle 1949, 244), which would limit its use in serving educational purpose. Metacognition is defined as consisting of 'metacognitive knowledge' (Flavell 1979, 1999): 'understanding how we think and learn and "metacognitive skill" (Veenman, Wilhelm, and Beishuizen 2004): the ability to create strategies which support our thinking and learning'. The models are proposed to enable discussion and advocacy for supporting imagination pedagogically, through their contribution to ever-transforming language and theory, conceived as emergent phenomena. They offer potential for explorative application in learning contexts, ultimately in support of increased democratic capacity (Dewey 1927, 147), largely via their metacognitive application (3). As a theoretical paper, this does not yet offer worked examples of the models in practice but provides an invitation to apply those theories.

From a pragmatic stance (e.g., Biesta and Burbules 2003), in order to address a problem which arose within my education practice: that an articulated understanding of 'imagination' is largely absent, I undertook a theoretical journey towards emergent models of imagination within cognition and in metacognition. According to Ryle ' ... there exists a quite general tendency among theorists and laymen alike to ascribe some sort of other-worldly reality to the imaginary' (1949, 232). Imagination is frequently regarded as a magical or frivolous escape from reality, as opposed to a fundamental aspect of our thinking, including but not exclusive to creativity. 'Imagination is magical. Take a moment to watch a young child play alone and you will experience ... the magic that comes from imagination' (Serf-Walls 2017). As a manager for the Creative Partnerships programme (2002–2011), our CEO asked me to 'look for the magic' resulting from the work, with this consisting of imagination. Many teachers told me 'it's all about imagination'. In 'Hard Times', Dickens suggests that, when regarded as 'fancy', imagination can be understood as oppositional to education focussed on accountable 'facts' and economic purpose. I have no desire to extinguish ideas of magical imagination but am mindful that a lack of definition could restrict the nurturing of imagination in our educational environment, where accountability can dominate (Clarke and Phelan 2015).

Imaginative escapism can offer psychological relief in a context of climate crisis, pandemic and the consolidated power of the mass media,[1] which enables the political manipulation of populations which may be ill equipped in terms of 'democratic capacity' (Dewey 1927, 147).

In a time of 'post-truth' (2016 Oxford English Dictionary) and 'fake news', it is easy to identify with dystopian fictions, which illuminate potential for authoritarianism by repressing imaginative thought and expression via 'newspeak' (Orwell [[1949] 1984] 1984) or by banning access to literature (Attwood 1985). Writers and artists (see Titchner, [Tate 2006]) excel in applying imagination in works which prompt us to question authority, exercising and provoking 'democratic agency'. If we accept that some artists (as imaginative 'experts') explicitly model processes in which imagination enables criticality and generates ideas for alternative possibilities which can empower

democratic agency, we can speculate that by supporting critical and possibility thinking, imagination is required for democratic agency generally. In this case we must support imagination to support democratic agency, defined here as 'having a responsible share according to capacity in forming and directing the groups to which one belongs' (Dewey 1927, 147).

Congruent with a perspective that 'Education as a democratic project is utopian in its goal of expanding and deepening the ideological and material conditions that make a democracy possible' (Giroux 2014, no page), a utopian school would seek to enable democratic capacity and therefore support imagination, ethically and pedagogically. This school would ensure opportunities to arrive at unexpected learning outcomes and gain confidence in undertaking that more open-ended process, partially through metacognitive understanding. Learners would be required to take ownership of their learning and be enabled to create knowledge (Jeffrey and Craft 2006, 47). With non-predetermined and personal outcomes, there would be potential to produce new ideas and transform understandings, enabling positive change.

This style of education has, to some extent, been realised through creativity programmes which strive for 'whole school change' (Thomson 2010) and through arts-based approaches focused on teacher development (e.g., the Paul Hamlyn Foundation Teacher Development Programme). Such approaches support teachers' interests in working within regulatory and audit frameworks to support creativity and imagination. This support could be increased.

In this paper, theoretical models of imagination in cognition and metacognition are offered as guiding structures to use in conceptualising and reflecting on pedagogy for imagination. Whether or not we strive for democratic capacity as an educational goal, imagination requires our attention as a key, psychological capacity for envisioning possibilities, in learners and for education 'as an emergent entity that does not simply serve a purpose, but also brings with it the purpose it serves' (Osberg and Biesta, 2).

Below, an overview of literature related to cognitive and metacognitive perspectives of imagination is followed by the methodology applied towards arriving at cognitive and metacognitive models. Two foundational concepts: 'Transcendence' and 'Cognitive Imagination' are highlighted, prior to the presentation of a model of imagination in cognition. The implications of this model are related to metacognition. The paper culminates with a model of metacognitive imagination and discussion of the possibilities for the models in education research and practice.

Overview of the literature: cognitive and metacognitive imagination

To overcome polarising or discipline-limited approaches to imagination and to ensure that new models are robust, a pragmatic, interdisciplinary approach was applied within an initial review literature relating to imagination, cognition and metacognition. The integration of conceptual data from different fields of knowledge was conceived to enable a novel contribution to education practice in being informed by understandings from beyond that practice. Exploring theories across disciplines provided ideas to use in *re-imagining* imagination, and enabled a conceptual escape from an educational perspective which can lead to pedagogy as a mechanism 'to facilitate a "ready-made" political or moral (normative) purpose' (Osberg and Biesta, 2).

What follows is an inexhaustive overview of multi-disciplinary literature, used to establish understanding of coverage, gaps and potentially fruitful areas in which to search for further sources.

Cognitive perspectives which include imagination in the context of art education have been explored by Efland (2002), Eisner (2002) and Arnheim (1969). The focus is not exclusively on imagination, rather; this is seen as an ingredient within creative cognition.

Ryle (1949), Warnock (1976) and Kearney (1984) have examined philosophical perspectives of imagination. Key philosophical theories related to the development of a cognitive perspective can be found in Aristotle's 'De Anima', Kant's 'Critique of judgment' and Hume's 'Treatise on Human Nature'. Piaget (1926, 1962, 1971) and Vygotsky (1986, 2004) provide developmental psychology perspectives. Robust investigations of mental imagery are contributed by Kosslyn (2005) and Paivio (2006). Examples of contemporary, psychological research into the role of imagination includes Byrne (2007), and Machin and Davies (2003).

Creativity research which includes imagination is relevant for considering imagination as cognition (e.g., Kaufman and Beghetto 2009; Stokes 2014), however; the concept of imagination itself is often left relatively unexpanded within discussions of creative cognition (e.g., Robinson 2001; Eisner 2002). Possibility thinking, as an aspect of imagination, is discussed and modelled by Cremin, Chappell, and Craft (2013), Craft et al. (2013) and Clack (2017). In the context of drama education Mark Cremin (1998) has explored modes of imagination coming closer to developing a way to analyse entire imaginative processes but acknowledging that the elaboration of these terms is minimal.

To consider *meta*cognition (e.g., Flavell 1979, 1999; Veenman, Wilhelm, and Beishuizen 2004; Tarricone 2011) in relation to imagination, understanding it as a kind of self-awareness applied to personal thinking and learning, it makes sense to first acknowledge imagination as a critical factor in species-wide human development which is associated with the emergence of consciousness of our own existence (e.g., Mithen 2001; Montell 2002; Donald 1991). Limited literature discusses the relationship between imagination and metacognition, excepting a light exploration through a focus on creativity (e.g., Runco 2015; Kaufman and Beghetto 2013).

Previous authors have described multiple definitions of metacognition (e.g., Tarricone 2011). The scope of this paper requires a pragmatic, essential yet comprehensive definition, to provide a foundation for relating it to imagination. According to Flavell: 'Metacognition' refers to one's knowledge concerning one's own cognitive processes and products or anything related to them.' (1979, 232). Flavell produced a framework including metacognitive knowledge, experiences, goals and actions (1979, 906). Meta-evidence demonstrates that supporting the development of metacognition in children increases their chances of educational success (Higgins et al. 2013). A strong link between imagination and metacognition implies that developing imagination is likely to be advantageous for learning more generally. If the relationship is strong and understood, we might better develop metacognition by better developing imagination, thereby improving individual learning, especially the capacity for self-regulation. This could equate to autonomy (Bonnett in Bridges 1997, 156), leading to increased potential for democratic capacity, as described by Dewey (1927) and Greene (2000).

Methodology

While few would argue against nurturing imagination, many practitioners might struggle to define it cognitively and support it accordingly. A pragmatic consideration of multi-disciplinary perspectives towards the emergence of a scaffold for thinking, dialogue and practice related to imagination, should help address these problems. We need to temporarily put aside reverential perceptions that imagination is a kind of 'magical' quality within our learning, making it seem unteachable, to explore it as a fundamental aspect of our intellect. To ensure the theoretical and material conditions for its growth, we need a way to understand and discuss it. One way to do this is by creating models (Moseley et al. 2005) of imagination, related to how we think and learn. A model, 'behaves in some way like the phenomenon it represents' (Moseley et al. 2005, 40), demonstrating a pool of relevant concepts and the relationships between them. A rigid model could lend itself to instrumental applications, predicting and applying causes to lead to effects, thereby denying opportunities for new or unprescribed learning. The models within this paper are not ultimate, rather, they are intended to provoke discussion and adaptation, as dialogical tools to encourage efforts to nurture imagination. The models in this paper aspire to Osberg and Biesta's suggestion to theorise education as 'emergent', requiring us to recognise ' . . . its continued "coming into being" and . . . open-ended purposiveness' (2018, 2).

Research consisted of purposively selecting and analysing multi-disciplinary literature relating to imagination, in order to support the emergence of models of imagination in cognition and in metacognition, for use in education practice. Following an initial literature review, a pragmatic, theoretical enquiry drew on diverse literatures related to historical (evolutionary and philosophical), psychological and arts perspectives. Rigour, as 'the extent to which a piece of research meets the requirements of its paradigm' (Baumfield, Hall, and Wall 2013, 26) was ensured by the regular comparison of theories and their implications related to practice. Incorporation of understandings from beyond the discipline of education enabled the emergence of interpretations within and across paradigms through a reflective enquiry process, using the literature as data.

Rather than studying 'relationships between language-in-use and the social world' (Johnson and McLean 2020, 377) which would constitute discourse analysis, analysis was conducted on a conceptual level, focused on theoretical ideas within the texts. A qualitative method, analogous to 'grounded theory' (Cohen, Manion, and Morrison 2008, 491), was applied inductively to the selection and analysis of literature. Theory emerged from the identification and cohering of themes from the data, in conjunction with the interpretive capacity of the researcher, as an education practitioner. Core, emergent categories were then expressed as narrative discussion.

Pragmatism has been accused of a lack of criticality through a perception that it tries to please everyone by incorporating every perspective (Crotty 1998, 62&73). In analysis and write-up, criticality was ensured through rich discussion, comparing theoretical assertions to personal experiences and across paradigmatic categories, in order to explicitly reject ideas which were not robust in the context of these combined lenses.

Foundational themes

Two themes emerged strongly. The first, *transcendence*, is concerned with imagination and self-awareness, in a species-wide and individual sense. The second is *cognitive imagination*, which is concerned with imagination as fundamental to thought.

From a species-wide perspective, the evolution of sufficiently sophisticated imagination has enabled our transcendental 'vision'. Evolutionary psychologists and archaeologists generally accept that human brains evolved and enlarged at a prodigious rate about 1 million years ago (e.g., Mithen 2001, 48), enabling the imaginative generation of self-concepts. 'This perception of one's individual existence in space and time … would become a driving force in the evolution of the human animal'. (Montell 2002, 8). Assuming that in enabling us to understand ourselves, it enables us to imagine the thoughts and feelings of others and to develop empathy and understanding of the needs of a social group, we can consider this developed capacity for self-awareness as a basic condition for democratic capacity. For example, self-awareness of our personal relationships might lead us to recognise that the holistic well-being of a social group depends on looking after its more vulnerable members and those close to them by empathising with their emotional needs.

Montell provides a convincing argument (2002, 17) for the evolutionary adaptation of imagination as driven, beyond the point in our imaginative evolution where we began to imagine the meaning of death, by an overwhelming fear of this (Mithen 2001, 49–50), which may have compelled us ' … to externalise this in the form of art and culture'. (Mithen 2001, 49–50). The resulting artefacts acted as cognitive anchors for concepts, so that within their presence, 'ideas could be recalled, manipulated and shared' (Mithen 2001). The materialisation of ideas as physical objects allowed us to be in their presence without having to hold them in our minds, freeing up 'space' to formulate, manipulate and share our thoughts, enabling us to over-ride existing mental categories and to progress our ideas, socio-culturally (Donald 1991). We see an iterative cycle, where evolved imagination leads to self-awareness, leading to socio-cultural dialogue, which further expands social and individual imagination. Applying this hypothesis to a current education context, if we see our theoretical models of imagination as cultural artefacts, introduced to teachers and practice, we could create cultural dialogue about imagination which expands social and individual imagination (about imagination). While imagination need not be applied explicitly in support of democratic agency, it is fundamental for developing this because it allows us to imagine ourselves, others and possibilities for existence/co-existence. As skilled 'imaginers', we develop capacity to explore and instigate democratic possibilities.

From a philosophical perspective; Kant shows us transcendence via imagination, occurring at the level of the individual, in the 'nitty gritty' of our thought. Kant's ideas are concordant with the evolutionary perspective in perceiving imagination as essential in elevating, or even permitting, our capacity to think. His theory of 'transcendental imagination' asserts that 'The principle of the necessary unity of pure (productive) synthesis of imagination, *prior to apperception*, is the ground of the possibility of all knowledge'. (Kant 2004, A118). Imagination is transcendental in that it precedes and is necessary for the rest of our thought. Transcendental imagination is an innate human faculty, underpinning our cognition by enabling us to relate and synthesise new, sensory impressions with pre-

integrated mental images. These have arisen from sensory impressions, organised into categories and bound together in our minds as concepts, by imagination. Transcendental imagination is a classifying agent, constructing categories based on sensed experiences, then adding to or re-grouping these when new sense experiences require integration. Imagination can be regarded as a dynamic, cognitive force, manipulating new sense perceptions and stored images to progress our understanding. This expands on the Aristotelian idea of imagination as the 'movement' of stored mental images, driven by 'desire' for meaning (Aristotle 1986, 3, 10, 433a). By establishing a coherence amongst our perceptions, we are able to construct understandings which allow us to rise above what would otherwise be a chaos of dis-ordered experiences, with this constituting a form of transcendence.

Without transcendental imagination, we couldn't synthesise our perceptions into concepts. If we experienced trees, we couldn't conceptualise them as part of the environment or consider that, in greater number, they would constitute a forest. We couldn't apply the concept of photosynthesis, or compare trees with other plants. Each experience of a tree would be unconnected with any idea which enabled a broader 'sense-making' about them and their place in the world. Likewise, without transcendental imagination about ourselves as humans, we couldn't understand the relationships between self, humanity and the universe. This transcendental deficit would negate democratic agency, since we could not understand the nature of relationships inherent in the concept of democracy or visualise a way to act in support of it. Kant's idea of imagination as a transcendental force is key to enabling democratic agency.

Transcendence implies the necessity for imagination in cognition and also in meta-cognition, where it enables a kind of 'second-level' transcendence. In metacognition, while focused on our thinking and learning, we consciously consider our 'first-level' acts of transcendence (occurring within our experiences of learning as knowledge formation). Imagination removes us from temporal, spatial and bodily constraints, freeing us to manipulate concepts, from the metaphorical 'outside'. It also enables us to think *new* thoughts, essential in any attempt to understand our unique capacities for and approaches to learning. An example is that of cohering an understanding of what helps us to remember things and thereby producing a strategy to improve our learning. There are synergies between imagination-enabled transcendence at this individualistic level and at the human-wide scale, for instance, in developing cultural artefacts to represent concepts, allowing us to access different perspectives on experience and use these to develop, socio-culturally.

By considering evolutionary and philosophical perspectives of transcendence through imagination, imagination can be understood as the means of becoming self-aware and creating meaningful concepts, including those about ourselves, which allow us to make judgements and take action. Imagination enables self-awareness by allowing us to consider ourselves as beings in space and time (past and future), in various situations, faced with real and hypothetical choices and their accompanying emotions. Self-awareness of our situations can enable us to imagine and empathise with those of others, who we can conceptualise as fellow humans. This empathy enables us to act critically, in consideration of other people. By enabling empathetic judgment and the envisioning of potential social action, imagination creates a route to democratic capacity.

Moving to the second emergent theme of *cognitive imagination*, we turn to Aristotle. By understanding imagination as a cognitive faculty we establish its place in education. In 'De Anima', Aristotle provides foundational ideas of cognitive imagination, defining it as a mediator between sense perception and judgment, necessary for logic and reason. It is the basis for consciousness because without it, our sense perceptions would not be accessible to our internal thought process. Mental images stand in for direct sensory perceptions:

> The thinking faculty, then, thinks the forms in images, and, as what it should pursue or avoid is defined in the images, it is moved even in the absence of perception, whenever there are images before it. (Aristotle 1986, *3, 4, 431b).*

Imagination is modelled as 'movement' (Aristotle 1986, 3, 3, 428b). Initially, this consists of moving sensory perceptions into the cognitive realm and storing them there for future use. The images support acts of deliberation and self-regulation, in which they are recalled and manipulated to support thinking. This is driven by a 'desire' (Aristotle 1986, 3, 7, 431b) to understand, to know or to make meaning. Thus driven, imagination connects sensory perception with our mental worlds, regulates our behaviour by enabling us to make choices and allows us to think backwards and forwards through time (Aristotle 1986). Humans have the ability to deliberately combine images, drawn from an internal store, in order to support reasoning (Aristotle 1986, 3, 4 429a., 3, 10, 433a-b). Imagination is ' . . . an intermediary faculty residing . . . *between* . . . faculties of sensation and reason'. (Kearney 1988, 112). The mental image, as a kind of building block of the imagination plays a critical role in cognition. Aristotle's theory is the starting point for building our models.

Aristotle instigated a cognitive perspective of imagination, most notably built on by Hume and Kant. Their ideas lay foundations for modern, psychological perspectives which directly influence contemporary education practice. Hume's [1739] 1985) almost existential concepts of how imagination enables us to hold together a world view by 'filling in the gaps' when empirical experience is not available resonates with Piaget's understanding of young children's imagination, which is required to compensate for a deficit in material experiences (Zittoun and Cercia 2013). Kant's concept of transcendental imagination, as an ordering force within our thinking, implies that imagination enables the construction of knowledge. The identification of imagination's role in cognitive ordering resonates with Piaget's theory of schema and the accommodation and assimilation of experiences into these (1952).

For Piaget, we construct a mental model of our world by drawing upon a store of mental representations related to sensory perceptions (Piaget 1971). Vygotsky agrees with Piaget in terms of how we accumulate, store, produce and use images, though he conceptualises this as a constructive activity rather than one of 'filling in gaps' (Zittoun and Cerchia 2013). This is particularly relevant in relation to the idea that imagination enables democratic capacity since it implies an empowered role for the learner as the creator of knowledge as opposed to an absorber of pre-existent truths. A further important implication is that it is possible to create *new* knowledge which provokes thinking about new possibilities.

Imagination enables us to create Vygotsky's zone of proximal development (Vygotsky 1986, 15) by constructing an idea of what learning may be achieved for ourselves or others with support. In playing this fundamental role within the growth of our thinking,

imagination largely enables our cognitive development. For Piaget and Vygotsky, the need to address disequilibrium within our thoughts activates imagination which is generated through sensory experience. When we encounter disequilibrium, according to Piaget, 'syncretism' (2002, 159–160) (connecting ideas) will be activated as part of an oscillating process whereby we extend our existing schemas or create new ones in order to accommodate new information.

The idea that imagination helps us to cope with consciousness and enables us to develop a coherent mental world, rising above the potential chaos of disordered, sensory experience, returns us to the theme of transcendence. Hume, Kant, Piaget and Vygotsky reveal cognitive perspectives which illuminate *how* imagination helps us to transcend ourselves. It helps us to cope with the potentially overwhelming experience of existence as a conscious being, necessary for survival in a somewhat 'chicken and egg' situation, where it is responsible for generating consciousness in the first place. Imagination emerges as the keystone of human stability, development and the means by which we hold ourselves and our society together.

An emerging model of cognitive imagination

In his 'Critique of Practical Reason', Kant provides a taxonomy of imagination which is (arguably) unsurpassed. Related to the rest of the literature explored, this has contributed significantly to the emergence of the models of imagination below. There are four major stages of imagination according to Kant (Johnson 1987, 147–165), summarised here:

1. Reproductive imagination: evoking mental images.
2. Productive imagination: connecting mental images to achieve a coherent and meaningful experience (synthesis).
3. Schematism: mediating between sensation and understanding/thought (ordering).
4. Creative imagination in reflective judgment: reflecting on representations in search of novel orderings (creativity).

These kinds of imagination call to mind frameworks of general, cognitive capacities (e.g., Bloom et al. 1956), implying that 'Imagination generates much of the connecting structure by which we have coherent, significant experience, cognition and language' (Johnson 1987, 165).

Kant's 'transcendental imagination' enables us to order our thoughts, to make sense of or to construct knowledge about our experiences. His taxonomy demonstrates this cognitive process of knowledge construction. In establishing a model of cognitive imagination for use in education contexts, it is practical to adapt Kant's model. Potential contexts for the application of the model are not limited to the arts and include humanities and STEM subjects across primary and secondary education, as well as learning in informal settings such as museums and galleries. The associations between imagination, metacognition and personal/democratic agency discussed through this paper suggest that developing imagination in education is worth pursuing across disciplines, in order to support learner agency. The models could be applied in interdisciplinary research which integrates opportunities for participant

learning where imagination is of interest, perhaps in relation to addressing global challenges, supporting innovation or developing democratic agency. While Kant's basic distinctions seem sensible, a model for use in such settings requires accessible language which very clearly distinguishes aspects of imagination. The model described below also incorporates Vygotskian, Piagetian and other concepts.

Within the model, 'Reproductive Imagination' becomes '*Imaging*', to negate confusion over the association of memory with recall, with memory more complex than the reproduction of a mental image. 'Productive Imagination' and 'Schematism' are unified as '*Productive Imagination*' (incorporating Piaget's 'syncretism'), to include the way in which we associate and order the images we have stored in our minds. 'Synthesising' fits into the '*Productive*' category but is also included in a new '*Creative Imagination*' category when we order or cohere imagery in any way which is personally novel. 'Creative Imagination in Reflective judgment' becomes '*Creative Imagination*', which includes reflection (using imagery) in order to arrive at original ideas. I add a final category to accommodate imagining which takes us beyond the usual 'laws' of material experience: '*Fantastical Imagination*'.

The model of imagination in cognition depends on accepting that imagination consists of various manipulations of mental images, recalled in order to cohere or create new ideas.[2] At its most basic, imagination consists of *'Imaging'*: evoking images of what is not present, to support our thinking. We have a kind of store of images to draw from (the 'allusionary base' [Broudy 1987, 18]). This is akin to pulling a document from a file to assist with a piece of work. We may then combine different documents or files towards a purpose: *'Productive Imagination'*. We may need or choose to combine or manipulate these images in order to think of something new, or at least new to us. Due to this *relative* originality, cited frequently as essential for creativity (see Kaufman and Beghetto 2009; Robinson 2001; Wegerif 2010), with Craft (2001, 56) describing a 'spectrum of novelty or innovation' as a necessary quality spanning high and 'Little c creativity'; I call this third type '*Creative Imagination*'. A step further is to enter the world of fantasy, to imagine things never *directly*, materially experienced: '*Fantastical Imagination*'.

In this model Creative Imagination is defined by originality. To be creative, a person needs to see ' . . . analogies between ideas that have not been previously related' (Robinson 2001, 135). These connections might be new only to the person doing the thinking or may be more generally original. The necessity to connect mental images towards an original idea means that Creative Imagination requires Imaging and Productive Imagination. It may also include Fantastical Imagination in arriving at a novel idea. In this model I am concerned with imagination's relationship with creativity as opposed to creativity itself, so imagination resides in the individual and serves creativity (Cremin 1998, 8).

Fantastical Imagination may or may not be creative. For instance, it could be said that bringing to mind a unicorn requires fantastical imagination since a unicorn is not 'real' in a material sense. This is not necessarily Creative Imagination because unicorns will usually not be a relatively original imagining. On the other hand, Fantastical Imagination could include Creative Imagination if the impossible scenario being imagined is original to the person imagining it. Fantastical Imagination, while it may or may not involve Creative Imagination, like Creative Imagination, will also rely on Imaging and Productive Imagination.

To consider how this model might apply in our thinking, we can anchor the categories to an idea and apply them in the context of a child learning. Let us use the idea of 'a mountain':

Imaging: they bring to mind an image of a mountain which they have experienced, in person or as a sensory representation. *Productive*: they consider and connect physical attributes from a variety of mountains which they have experienced. *Creative*: they produce an idea of a mountain (which could theoretically, exist materially) but which they have never experienced before. *Fantastical*: they bring to mind the 'Lonely Mountain', complete with the dragon, Smaug, from 'The Hobbit'. Note that fantastical imagination is 'fantastical' because the child imagined something materially impossible, yet not original.

An emerging model of metacognitive imagination

Evidence demonstrates that the development of metacognition in children supports their achievement and attainment in school (Higgins et al. 2013) implying that, if imagination is important within metacognitive processes, we can support children's broader learning by nurturing it. Given the fundamental role of imagination within cognition, we can assume that imagination supports metacognition (Flavell 1979). With 'meta' meaning 'referring to itself' (Cambridge dictionary) *meta*cognition includes imagination because it requires the application of our cognition to a focus on our cognition. In metacognition, the types of imagination identified in the cognitive model would be focused on our own thinking and learning.

Simplistically, modelling imagination and metacognition requires recalling the cognitive model and objectifying it to gain a meta-perspective. While imagination in our metacognition is likely to function in the same way as in our cognition (focused on our own learning) the relationship is more complex, particularly regarding a greater role for self-awareness, as dependent on imagination. Imagine a child, metacognitively reflecting on how they produced a story about the future of the planet. The types of imagining are interacting within a self-transcendent lens, as the child considers how they incorporated experiences from a visit to the zoo to create their vision of a world run by talking animals. Via imaginative self-awareness, the child arrives at metacognitive understanding that drawing on personal experience inspires their creativity. Metacognitive skill develops as they decide on this as a future learning strategy. They use their cognitive capacities in the context of their transcendental capacity for self-awareness, with both capacities supported by imagination, within metacognition.

Self-awareness seems necessary for metacognition. Above, I discussed Kant's idea of transcendental imagination in relation to developing coherent concepts of 'first-level' acts. Proust describes a function of metacognition as ' . . . that of applying . . . concepts to first-order contents, which in turn provides the inferential structure needed for reasoning about one's own states . . . ' (2007, 295). This can be related to Productive and Creative Imagination. Productive Imagination coheres imagery of experience to imagery of concepts related to thinking and learning. Creative Imagination applies relevant concepts in order to reason about one's own states – necessarily a relatively novel piece of thinking.[3] The implication is that representation of the self from the inside surely requires imagination as I have defined it.

If self-awareness is the foundation of metacognitive knowledge, it is reasonable to expect imagination to have an amplified role in metacognition. This deliberate application of our thought towards our own thinking also implies the use of Creative Imagination. It is as if, when breaking through a barrier between cognition and metacognition, the imaginational bar is raised. Rather than beginning with the 'basics' of imaging and productive thought, entering into metacognition initially demands Creative Imagination. Imaging and Productive Imagination are used but metacognition is based on cognitions already acquired through socio-cultural experience and previously turned into images, using Productive and Creative Imagination at the cognitive level. These are now material for metacognitive consideration as well as tools for metacognitive manipulation – the same thought processes applied within a focus on learning. Only when we have embarked on the creative task of developing metacognitive knowledge and skill do we continue to manipulate and combine imaging and products of imaging at this higher level, 'looking down' on our own learning, having accumulated a sufficiently sophisticated lexicon of symbolic language in order to do so. In Vygotskian terms, we have crystallised our thoughts into mental tools which enable us to unlock and work within a metacognitive realm.

As an example of applying the metacognitive model to a piece of thinking, we will use the example of a child faced with a challenging exam. *Imaging*: they bring to mind an experience of being unable to remember information required in an exam situation. *Productive*: they recall remembering information after writing it down and remembering information presented as a mind map by a teacher. *Creative*: they reflect that combining writing with images aids memory. For the first time, they make their own mind map relating to the exam. *Fantastical*: they imagine an infinite gallery of mind maps about every possible topic, including those which people haven't even thought of yet. The idea of this collection inspires them to produce further maps, as a strategy for memory.

In this example, Fantastical imagination is also Creative, as the learner has not previously experienced or thought of an impossible, infinite library. Imaging and Productive imagination overlap, as Productive imagination needs images to connect. Thoughts become metacognitively skilful at the Creative stage, when the learner makes their mind map and again, following the Fantastical stage, when they recognise their strategy. On arriving at a strategy, they can plan their learning, with autonomy. In supporting autonomy this metacognitive skill has potential for enabling democratic capacity. Pedagogies which nurture imagination towards supporting metacognition could enable learners to develop personal and democratic agency.

The example above is simplified. While the categories will interplay more it provides a starting point for considering what are often tacit processes. Examining how kinds of imagination map onto types of metacognition; metacognitive knowledge and skill can be situated within a larger 'pool' of Imaging in relation to thinking and learning, from which we can bring to mind what is not present in relation to 'learning'. Ready-cohered within this pool of images is a range of cognitive tools which, when directed towards the concept of our own learning, enable us to manipulate and generate new thoughts which enhance metacognitive knowledge and skill.

Metacognitive knowledge and skill rely on and feed each other, with skill requiring knowledge as a basis from which to construct successful strategy and the obtainment of knowledge requiring the development of skill. Metacognitive knowledge requires

Productive Imagination in order to recall existing and integrate new images, but Creative Imagination is needed in order to synthesise these thoughts by making and applying 'rules' to organise them coherently. It might seem counter-intuitive to include Creative Imagination within metacognitive knowledge, seeing the latter as an established mental resource but this author has an ontological perspective that knowledge is self-constructed; made by combining experience and established thought. This requires Creative Imagination to produce relatively new, valuable knowledge. Thus, I suggest that metacognitive knowledge is supported by both Productive and Creative Imagination, (without interrogating whether metacognition is necessarily 'new'), speculating a weighting towards Productive Imagination in terms of frequency.

Now considering metacognitive skill, with its focus on self-regulation and strategy, surely this is largely supported by Creative Imagination, since these executive functions require the ability to reason and solve problems which are personal (therefore novel) to the learner. While it is likely to draw directly on Productive Imagination (and imaging), in order to enable connections to be forged and thoughts to flow, it also draws on Productive Imagination already crystallised into metacognitive knowledge as imagery related to learning. It requires the ability to hypothesise and to create strategies for application in a range of contexts which may require adaptation. Almost all definitions of intelligence specify that it 'involves the ability to adapt to the environment' (Sternberg 2019, 1). This metacognitive capacity for 'adaptive intelligence' (Sternberg 2019), applied to our own learning, is essential for re-positioning ourselves as learners within and responding effectively to ever-transforming, socio-cultural environments. It can enable us to feel confident as democratic agents in the face of challenge.

It is difficult to conceive of much of metacognitive skill as not requiring imagination if 'Task analysis, planning, monitoring, checking, and recapitulation' (Veenman, Wilhelm, and Beishuizen 2004, 90) constitute this; however, we should ask whether Creative Imagination is still present if metacognitive skilfulness is employed unconsciously, when it has been pre-developed. Metacognitive skills ' . . . can be acquired and eventually executed implicitly' (Veenman, Wilhelm, and Beishuizen 2004, 90). The boundaries between cognition and metacognition blur here. In situating imagination with metacognitive skill, we can say that we are left with Creative Imagination underpinning the majority of the latter, leaving space outside of the imaginative aspects of metacognitive skill to account for non-creative (non-original) execution of existing strategies and acknowledging the unaddressed question of whether creative imagination is necessarily conscious.

Puryear (2014) and Runco (2015) question whether metacognition and creativity/imagination are so interrelated that they are the same thing. Having defined Creative Imagination as imagination applied relatively originally, if I also include the condition that it should produce a valuable outcome, this implies the need for a meta-perspective (metacognitive knowledge, supported by Creative Imagination) of our own learning, to which we can apply creative imagination, to produce that outcome. This outcome might be classified as metacognitive skill or alternately, new metacognitive knowledge. We need to 'rise above' what exists, to see beyond and construct the new, based on self-knowledge.

> An individual's knowledge of self … is the ultimate medium of creative behaviour, for knowledge determines decisions as much as opportunities. In fact, it is the basis of one's knowledge that one can perceive and identify one's opportunities (Albert 1990, 19)

Combined with the idea (above) that knowledge and self-awareness develop transcendentally as a result of imagination, Albert's argument suggests that imagination, by enabling self-awareness/knowledge, is necessary for the development of creativity. Presumably if we apply this creative thinking to our own learning then our self-awareness is nurtured further, generates improved learning and multiplies our creative potential in an on-going cycle.

The implication is that metacognition improves creativity. It is also argued (above) that Creative Imagination is necessary for metacognition. Perhaps metacognition, Creative Imagination and creativity are so closely and iteratively related that they are the same thing by different names? Resolving this issue hinges on the concept of originality and whether metacognition is 'original'. We might consider ourselves as having a permanent sense of metacognition, which, while ever-developing, contains knowledge of and strategies for our learning which remain constant, for example; I know that my memory isn't perfect and that it helps to write things down. This is not a new thought but still seems to qualify as metacognitive. For now, I settle on the idea that creativity and metacognition are closely intertwined, united in drawing heavily on all kinds of imagination, particularly; Creative Imagination. This suggests that imagination in all its forms is fundamental to and has a transformative impact within an iterative, self-developmental learning process which enables metacognition/creativity applied to our own learning. Imagination is essential for increasing our capacity to learn.

Now to situate Fantastical Imagination within the model. Initially, this category seems out of place in a model of thought which applies to the practical focus of 'learning'. However, the arguments posited in terms of Fantastical Imagination within cognition remain relevant to metacognition. When focusing on our learning, Fantastical Imagination might help us to consider possibilities which are far out from or impossible within our learning contexts but provide 'counterfactual' (Byrne 2007) criticality which enables us to arrive at realistic possibilities. A learner reflecting on their achievements in physics might ask themselves, 'what if I'd been able to talk with Stephen Hawking?' or ' what if I'd been ill and unable to attend lessons?' This counterfactual thinking could enable them to understand the realistic possibilities for their development: 'I have access to teachers and I need them to help me learn'.

Arguments which support the value of fantasy in terms of human evolution can be aligned with the individual learning process. If, as Mithen (2001) suggests, an evolutionary function of fantasy was to enable us to over-ride existing mental categories by externalising thoughts at odds with our innate sense of order, then fantasy might help us as individuals to arrive at new possibilities for our learning. For example; when previously metacognitive strategies become cognitive and we need new ways to understand our learning. Fantastical Imagination may also be required for Piaget's 'formal operations', requiring a high degree of abstract thought. In a metacognitive context, we might play with and hypothesise through abstracted or metaphoric ideas related to our thinking. We may conceptualise interactions between abstract, cognitive functions, such as 'understanding', 'applying' and 'creating' by relating these to fantastical contexts, for instance;

we might imagine the cognitive processes involved in understanding the fundamental nature of the universe and making a new one. Using fantastical metaphor, we might visualise the process of developing our cognition as building a magical castle and hypothesise that by working with others to add building blocks, we will make the castle/our cognition stronger. By considering extremes and manipulating metaphoric categories beyond our material realm, we could extend the boundaries of the metacognitive knowledge and skill we had previously arrived at. For these reasons I locate Fantastical Imagination at transitional areas of metacognition, or at the 'edges' of knowledge and skill, suggesting that it acts as a catalyst in the transformation of our learning.

The discussion above has led to the emergence of the model which is presented as a diagram below (Figure 1).

Figure 1 shows the types of imagination previously described, represented within metacognition. Metacognitive knowledge and skill interact with each other in a dynamic, two-way flow, dependent on and almost filled by imagination. Remaining within these processes, beyond imagination, are operations which do not require mental images (verbal coding and perhaps, structures for thinking through, *if* it is possible for these not to depend on imagery, which is shown as foundational in the model as a whole). In order to reach this metacognitive state, through Imaging, we use imagination to transcend into self-awareness (in a species-wide and individual sense), so that imagination is the basis for our knowledge construction. We then use (within our lived experience

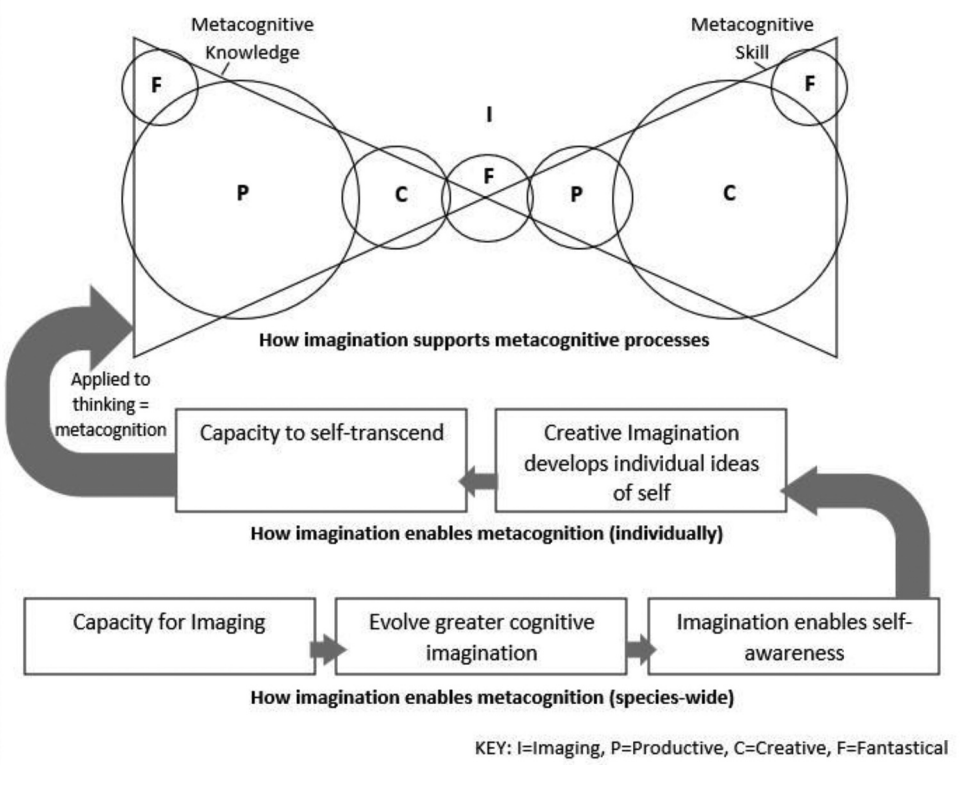


Figure 1. Model of imagination leading to and within metacognition.

rather than as an evolved biological feature) creative imagination to crystalise our concept of self, so that we are able to consider ourselves as if we are outside of our own cognition, transcending again to become metacognitive. Imagination then continues to support this metacognitive state.

Conclusion and next steps

The emergent models imply that imagination is fundamental to cognition, metacognition and is the root of agency. Nurturing imagination is critical for ensuring personal and democratic agency so should be an educational priority, effectively pedagogically supported.

To think, we must imagine. Imagination is not a magical addition to thought but intrinsic throughout. The necessity for imagination in metacognition is particularly important, given that metacognition supports achievement and attainment. By association with metacognition, imagination could share some of the credit for learning with the former attribute, raising the status of pedagogies for imagination and enabling educational opportunities for imaginative development.

As pragmatic and interdisciplinary, the emergent models offer a new, theoretical perspective, ready for adaptation to practice. In allowing the models to 'emerge' theoretically, they align with Osberg and Biesta's conception of education (2018, 2) 'as an emergent entity that does not simply serve a purpose, but also brings with it the purpose it serves'. By producing theoretical models of imagination we can imagine new possibilities for education, since the expansiveness of a theoretical emphasis, while providing touchstones, does not imply finality. Instead it provokes us to explore opportunities through its application in supporting learning and in further developing the theoretical models themselves. The theory which has emerged implies the critical importance of developing imagination within education, so bringing new purpose to its pursuit. In pursuing and prioritising imaginative development, there is potential to nurture educator capacity to re-imagine and contribute to a continued emergence of 'education'.

The emergent models provide two key opportunities. First; they offer a basis for dialogue. This may lead to their adaptation, rejection or replacement. There is no claim that they are comprehensive. They offer a means of discussing what has frequently been considered a largely indescribable, yet somehow valuable part of our learning. Secondly; by exploring their potential in practice and research for imagination, we can find out how useful they are.

The models can help us to establish the current state of practitioner understanding of imagination, the in/adequacy of support offered for this and provide a framework for pedagogic design. They could be used to form criteria for the exploration of imagination in and across contexts, in relation to multiple factors and impacts (not to suggest a solely deductive approach; the models might complement inductive methods). A priority is to expand understanding of the imaginative meta/cognitive processes which support capacity for personal and democratic agency.

The models may be particularly helpful for those working to develop creative thinking, often through the arts, as described at the outset of this paper. According to Eisner,

> Work in the arts . . . invites the development of a disposition to tolerate ambiguity, to explore what is uncertain, to exercise judgment free from prescriptive rules and procedures . . . Such a disposition is at the root of the development of individual autonomy. (2002, 10)

If the arts nurture autonomy (via imagination) more effectively than other subjects, there is an argument to raise their current status. They may provide a space to develop transferrable thinking and nurture thinkers who are capable of effecting positive, social change by being '. . . able to imagine something coming of their hopes' (Greene 1995, 25). Considering theories that producing art contributed to our evolution by enabling the cultural sharing of ideas, it follows that, more immediately, tailored art experiences could develop individual and socio-cultural imagination towards social progress. It would also be helpful to examine imaginative process in the humanities and literature, given that these subjects can inform the generation of personal and cultural imagination.

By revealing what are usually tacit, cognitive processes, applying the models could provide new understanding of pedagogic needs. Through application in practice and research, across subjects, adapted versions of the models may emerge and help us to effectively evidence, advocate and continually transform imagination in education, contributing to personal and democratic agency via multiple routes.

My pursuit of emergent models of imagination in cognition and metacognition was undertaken with the conviction that imagination contributes to a good life by adding pleasure but also by nurturing individual and democratic agency. The relationship between imagination and democratic capacity is strong but requires elaboration. The theoretical models I have arrived at suggest that imagination is the root of and is deeply integrated within processes in which it supports creativity which, in turn, supports metacognition. In enabling self-awareness and our capacity to think strategically about our learning, metacognition enables the autonomy required for democratic capacity. The question now is whether and how, equipped with models of imagination, these can support us to release imaginative education for imagination.

Notes

1. In the UK, 3 companies dominate 83% of national newspaper distribution (https://www.mediareform.org.uk/who-owns-the-uk-media), with 6 billionaires owning or having a majority of voting shares in most of the national newspapers (https://www.opendemocracy.net/en/opendemocracyuk/five-reasons-why-we-don-t-have-free-and-independent-press-in-uk-and-what-we-can-do-about/).
2. Discussion of well-trodden paths regarding the nature of mental imagery is limited by the scope of this paper.
3. Acknowledging arguments that metacognition may, in part, be unconscious.

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